The Fourbar Mechanism

A closed kinematic chain composed of 4 rigid bars/links connected in pairs using revolute/pin joints, with one of the links fixed in called the four-bar mechanism. These four links in the most general four-bar linkage are termed: the *ground link*, the *input link*, the *floating/coupler link*, and the *output link*.

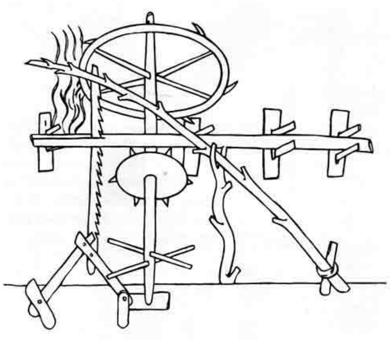
Figure 1: *A four bar linkage from the Reuleaux Kinematic Mechanisms Collection, Cornell University Library (visit: https://digital.library.cornell.edu/catalog/ss:372514)*

The four-bar linkage is the simplest and often times the most useful mechanism.

Applications

"*While I am not aware of unequivocal evidence of the existence of fourbar linkages before the 16th century (CE), their widespread application by that time indicates that they probably originated much earlier.*" – E. S. Ferguson, 1962.

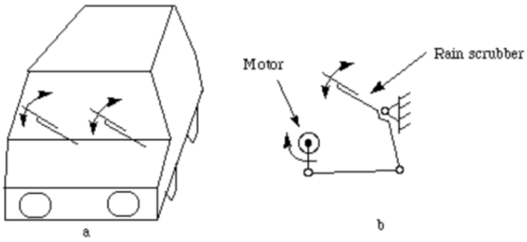
1. An early example of the use of the fourbar linkage is in an up-and-down sawmill of the 13th century. The guide mechanism at lower left, attached to the saw blade, appears to be a 4-bar linkage.



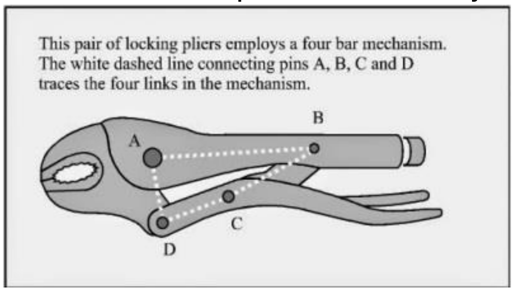
2. A ***pumpjack*** uses a fourbar mechanism to convert the rotary motion of the engine into reciprocating motion to pump underground oil to the surface.

3. A ***bicycle*** uses two cranks (the pedals) to convert the reciprocating motion of human legs into rotational motion.

4. ***Windshield wipers*** use fourbar linkages to convert rotational motion of the motor into oscillating motion of the wipers.



5. ***Locking pliers*** use a fourbar mechanism to open or close the jaws.



References

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